VocalEyes: A Context-Aware Real-Time Assistive Vision System with Priority-Based Audio Guidance

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***Abstract*—Visual impairment significantly limits the ability of an individual to interpret surrounding elements and navigate environments, thus affecting autonomy and quality of life. Although a range of assistive technologies is available, most are limited to primitive object detection or narration and therefore fail to provide context-aware, safety-related decision-making. This paper presents VocalEyes, a machine learning–based assistive system that converts real-time visual information into smart audio feedback for users with visual impairments. The architecture incorporates deep learning–based object detection, spatial estimation, priority classification, intelligent silence, and optical character recognition (OCR) into a coherent offline system. VocalEyes processes live camera streams, identifies objects, and estimates relative distance and direction through bounding-box analysis. A priority-based decision module classifies detected entities into safety-critical and non-critical categories, ensuring timely notification of hazardous situations while minimizing redundant feedback. An intelligent silence mechanism reduces repeated announcements, enhancing usability. OCR supports reading of environmental text such as signboards and documents, and is dynamically disabled when hazardous conditions occur to prioritize safety. The system supports live-camera operation, making it affordable to test and deploy. Real-time performance, accurate spatial perception, controlled speech output, and functional OCR are empirically evaluated and demonstrated to be reliable. Unlike traditional assistive systems, VocalEyes interprets context and makes decisions rather than merely narrating observations. The framework is entirely offline, cost-effective, and modular, making it suitable for resource-limited environments.**

***Index Terms*—assistive technology, visual impairment, object detection, priority-based alerting, OCR, intelligent silence, real-time audio guidance, context-aware systems**

I. Introduction

Visual impairment places significant barriers on independent living, as individuals cannot safely navigate environments, identify obstacles, or access visual information such as signboards and printed materials [8]. Traditional assistive technologies such as walking sticks and guide dogs offer only rudimentary mobility support without detailed context awareness. Consequently, blind individuals often rely heavily on others, reducing their independence and quality of life.

Recent advances in artificial intelligence—specifically in computer vision, deep learning, and speech synthesis—have made it feasible to create intelligent assistive technologies that interpret visual data and deliver auditory instructions [1]. Object detection models, OCR systems, and real-time image-processing pipelines have shown promising performance in bridging the perception gap for visually impaired users [10]. However, the majority of existing solutions focus on object recognition or text decoding in isolation, without consideration of environmental context, safety prioritization, or cognitive load management [9]. Most assistive systems synthesize continuous speech for every identified object, producing excessive audio feedback that reduces usability [4]. Furthermore, OCR modules typically operate regardless of environmental safety, potentially distracting the user during emergencies [3].

These limitations highlight the necessity of a more advanced assistive system that not only detects the surrounding world but also determines what information is relevant and when it should be communicated. This paper presents **VocalEyes**, a context-sensitive, real-time assistive vision system that provides intelligent audio guidance to visually impaired users. The proposed system integrates object detection, distance and direction estimation, priority-based alerting, intelligent silence control, and safety-controlled OCR into a single architecture. Unlike traditional detect-and-speak paradigms [1], [10], VocalEyes employs a decision-based pipeline that evaluates environmental conditions before generating speech output. By prioritizing safety-critical objects, minimizing unnecessary alerts, and enabling text recognition only under safe conditions, VocalEyes delivers guidance analogous to a human assistant rather than raw sensory data. The system operates entirely offline and supports live-camera mode, making it ideal for low-cost deployment.

A. Project Idea

The core concept of VocalEyes is the creation of a smart assistive vision system capable of converting real-time visual information into meaningful auditory instructions for visually impaired users. The system acts as a virtual guide by providing context-sensitive speech that helps users understand their environment. Live video is captured by a camera, and each frame is processed through a deep learning–based object detection model [1]. Detected entities are analyzed to approximate relative distance and direction. A priority engine then determines which objects warrant attention and which can be suppressed.

High priority is assigned to safety-critical elements such as vehicles, stairs, and trip hazards, while less important objects are suppressed unless they pose a risk. OCR is also integrated, allowing the system to read visible text such as

signboards and notices [3]. Unlike traditional OCR-based assistive systems [3], VocalEyes activates text reading only when the surrounding environment is determined to be safe. An intelligent silence mechanism recalls past alerts and refrains from repeating similar information unless there has been a change in distance, direction, or environmental context [9]. All processing is performed offline, enabling deployment on standard computing devices without internet connectivity [6].

B. Motivation

Despite impressive advances in AI and computer vision, visually impaired individuals still face considerable obstacles in independent movement and environmental awareness [8]. Traditional assistive tools such as canes and guide dogs do not provide adequate spatial information, object recognition, or contextual awareness. Consequently, reliance on other people persists, causing reduced confidence, restricted mobility, and increased safety risks.

A number of AI-based assistive systems have been proposed to address these problems [6], [7]; however, the vast majority focus only on object detection or text reading, disregarding practical usability [9]. Continuous narration of all detected objects generates excessive audio, which can overwhelm users [4]. Equally, OCR reading systems often operate independently of environmental context, creating a risk of distraction when navigating hazardous areas [3].

A second driving force is the scarcity of transparent and customizable decision logic in commercial assistive devices [2], [7]. Most commercial systems have black-box implementations in which alert generation and prioritization are unclear. Furthermore, many require cloud connectivity or dedicated hardware [5], [6], limiting accessibility in low-resource settings. The motivation behind VocalEyes is the development of a more intelligent, context-aware assistive technology that behaves like a human guide rather than a data-reciting machine. The system is designed to prioritize safety-critical objects, minimize redundant speech, dynamically control OCR, operate fully offline, provide distance and direction awareness, and reduce cognitive load. VocalEyes seeks to improve independence, safety, and user experience for visually impaired individuals through an affordable and practical solution.

C. Scope

The VocalEyes project aims to develop and deploy an AI-based real-time assistive vision system providing intelligent audio guidance to visually impaired users. The major components include: (1) real-time object detection using deep learning models to locate people, vehicles, stairs, furniture, and common obstacles [1], [10]; (2) bounding-box–based distance and direction estimation without additional depth sensors [5]; (3) priority-based decision logic to categorize objects by safety relevance; (4) intelligent silence features to reduce repetitive alerts [9]; (5) OCR for reading environmental text with safety-controlled activation [3]; (6) offline text-to-speech synthesis [6]; and (7) live-camera operational mode. The system runs on standard computing hardware without internet connectivity. GPS-based positioning, face recognition, and medical diagnostics are outside the current scope and are identified as potential future enhancements.

II. Related Work

Assistive technologies for visually impaired users have evolved significantly due to advances in AI, computer vision, and wearable computing. Early assistive devices relied on ultrasonic sensors to provide basic proximity alerts through audio or vibration [8], but these systems lacked semantic understanding and could not distinguish between critical and non-critical objects.

The advent of deep learning and convolutional neural networks (CNNs), together with object detectors such as YOLO, enabled real-time recognition of multiple objects in a scene [1], [10]. Several studies adopted camera-based detectors that translate detected object labels into speech output [7]. Although these systems improved environmental awareness, most followed a direct detection-and-speak approach, producing repetitive and mentally taxing narration in crowded environments [9].

Concurrently, OCR-based aids were developed to assist the blind in reading printed text [3]. These systems use image preprocessing and OCR engines to extract text and convert it to speech. However, OCR modules in current systems operate independently of environmental context and may continue reading text even in dangerous scenarios [3]. More sophisticated systems attempted scene-level descriptions using image captioning and natural language generation [4], providing detailed environmental overviews at the cost of verbose and non-prioritized narration unsuitable for real-time navigation.

Commercial wearable assistive devices [2], [7] have integrated object recognition and OCR, but these solutions are typically proprietary, costly, black-box systems with minimal decision logic transparency. Many also rely on cloud-based processing [6], introducing latency, privacy concerns, and offline limitations. Smartphone-based assistive applications [6] provide object recognition and text reading via cloud AI but suffer from internet dependency and privacy risks. RGB-D camera–based systems [5] improve spatial awareness through depth sensing but require costly additional hardware. Small object detection approaches such as YOLO NAS variants [9] improve indoor detection accuracy but still lack context-driven decision logic.

Based on the reviewed literature, the following research gaps are identified: (1) absence of priority-based alerting to differentiate high- and low-risk objects [1], [10]; (2) lack of intelligent silence features to avoid speech repetition [9]; (3) absence of integrated object detection and OCR under a unified safety framework [3]; (4) limited spatial awareness without additional depth sensors [5]; (5) excessive narration causing cognitive overload [4]; and (6) dependence on cloud infrastructure in most commercial solutions [6]. VocalEyes directly addresses these gaps by combining object detection, spatial estimation, intelligent silence, and safety-controlled OCR into a unified, decision-based framework.

III. Literature Survey

Table I summarizes representative work in the field of assistive vision systems for visually impaired users and identifies key research gaps addressed by VocalEyes.

A. Identified Research Gaps

Based on the reviewed literature [1]–[10], the following gaps are established:

- **Absence of context-sensitive decision making:** Most systems [1], [9], [10] transform detected objects directly into speech without analyzing significance, urgency, or relevance.
- **Lack of priority-based notification:** Current solutions [4], [8] do not differentiate between high-risk objects (vehicles, stairs) and low-risk objects (furniture).
- **No intelligent silence mechanism:** Monotonous announcements [9] induce cognitive overload and reduce usability.
- **Uncontrolled OCR activation:** Text-reading modules [3] operate continuously, even in hazardous situations.
- **High hardware or cloud dependency:** Many systems [5], [6] depend on costly hardware or cloud processing, restricting offline usage.
- **Poor distance and direction awareness:** Few systems [8], [10] estimate spatial positioning without additional sensors.

B. Positioning of the Proposed System

VocalEyes addresses the aforementioned gaps [1], [3], [5], [6], [9] through: context-based decision logic; priority-based object classification; bounding-box distance and direction estimation; intelligent silence; safety-controlled OCR activation; fully offline operation; low-cost hardware compatibility; and dual-mode operation (simulation and live camera).

C. Problem Statement

Existing assistive systems [1], [4], [9], [10] have low functional versatility because their implementations focus primarily on object detection or text recognition alone, following a direct detection-and-speech approach with continuous narration regardless of context or cognitive load. Users are frequently subjected to audio overload, loss of situational awareness, and inability to receive safety-critical information in a timely manner. OCR-based systems [3] are not subject to environmental awareness and can distract users during hazardous situations. Commercial devices [2], [5]–[7] tend to be costly, cloud-dependent, and lack transparent decision processes. There is thus a clear need for an intelligent, low-cost assistive vision system that detects objects, reads text, comprehends the surrounding environment, prioritizes safety, eliminates redundant information, and delivers useful auditory guidance in real time.

D. Objectives

The VocalEyes project objectives are to: (1) realize a real-time assistive vision system based on deep learning [1]; (2) effectively detect common objects and obstacles in live camera feeds while estimating relative distance and orientation [5]; (3) introduce a priority-based alert mechanism for safety-critical cases; (4) implement intelligent silence to reduce unnecessary audio prompts [9]; (5) integrate OCR for legible text extraction [3]; (6) establish safety-controlled OCR activation; (7) provide real-time offline text-to-speech feedback [6]; and (8) support live-camera trials and field implementation within a cost-effective, modular architecture.

IV. Case Studies

This section presents realistic scenarios illustrating how VocalEyes operates effectively. Each case study demonstrates the system's context-aware decision making, priority alerting, and intelligent silence mechanisms.

A. Case Study 1: Indoor Navigation

In an indoor setting such as an office or classroom, VocalEyes identifies surrounding people, furniture (chairs, tables), and doors using the object detection module [1], [9]. The system estimates the distance and direction of each detected object and delivers audio guidance only for relevant entities [5]. For example, when a person is within close range, VocalEyes announces their presence and direction. Furniture objects are suppressed until they fall directly in the user's path. Where screens or noticeboards are present, OCR is enabled only when no immediate obstacle is sensed [3]. Intelligent silence prevents repeated announcements when the spatial arrangement has not changed [9]. This case study demonstrates the system's ability to support safe indoor navigation with minimal unnecessary narration.

B. Case Study 2: Outdoor Navigation

In an outdoor environment, the system prioritizes safety-related items such as vehicles and stairs [8], [10]. When a vehicle enters close range, VocalEyes generates an alarm and disables OCR until the hazard has passed. Distance-based scaling makes alerts more urgent as proximity increases [5]. Direction awareness provides messages such as "vehicle approaching from the right," enhancing spatial knowledge [7]. This case study validates the effectiveness of priority-based alerting in dynamic outdoor environments.

V. Framework and Methodology

A. System Framework

VocalEyes is designed as a modular, decision-focused architecture that converts visual cues into meaningful audio instructions. The general pipeline is:

Camera Input $\rightarrow$ *Object Detection* $\rightarrow$ *Spatial Estimation* $\rightarrow$ *Priority Classification* $\rightarrow$ *Intelligent Silence* $\rightarrow$ *OCR (Conditional)* $\rightarrow$ *Text-to-Speech Output*

TABLE I
SUMMARY OF RELATED WORK AND RESEARCH GAPS

Ref.	Approach	Methodology	Contributions	Limitations	Research Gap
[1]	Real-Time Object Detection	YOLO/CNN detectors with audio feedback	Real-time obstacle detection for visually impaired	Direct object-to-speech; repetitive alerts; no spatial awareness	No intelligent decision-making or cognitive load reduction
[2]	Wearable Smart Glasses	Edge inference with embedded camera and speech output	Wearable assistive indoor guidance platform	No customizable prioritization; limited explainability	Need for user-adaptive priority and decision logic
[3]	OCR-Based Reading Aids	OCR with text-to-speech synthesis	Reading of printed materials and documents	No safety awareness; high false positives	No contextual safety integration with OCR
[4]	Scene Description Systems	CNN scene understanding with NLG	Full environmental descriptions	Verbose; unsuitable for real-time navigation; no prioritization	No selective, priority-based speech filtering
[5]	RGB-D Multimodal Devices	RGB-D cameras with visual positioning	Improved spatial awareness and indoor localization	High hardware cost; sensor dependency	Need for low-cost distance estimation without extra hardware
[6]	Smartphone Assistive Apps	Cloud-based AI with camera input	Object recognition and text reading	Internet dependency; latency; privacy risks	Need for offline, explainable assistive systems
[7]	Smart Glasses Prototypes	Edge inference with speech output	Wearable assistive vision platform	No custom prioritization; limited explainability	Need for user-adaptive priority logic
[8]	Audio Obstacle Alert Systems	Ultrasonic sensors with audio alerts	Basic proximity warnings	No object classification; no semantics	Requires semantic identification and contextual awareness
[9]	Small Object Detection	YOLO NAS with Super Gradients training	Improved indoor small object detection accuracy	No context-aware decision logic; no safety gating	Intelligent silence and safety-driven suppression required
[10]	YOLO-Based Assistive Vision	YOLO detection with text-to-speech feedback	Feasibility of AI-based assistive vision	No OCR; no priority logic; isolated subsystems	Need for integrated, context-aware, safety-driven architecture

The system operates in **Live Camera Mode**, supporting real-time processing via webcam or similar imaging device. All computation is localized to avoid cloud infrastructure [6], guaranteeing offline support and eliminating latency.

As illustrated in Fig. 1, the system begins with live camera input and frame acquisition. Captured frames are processed by a deep learning–based object detection module [1] to identify entities such as pedestrians, vehicles, animals, and obstacles. Bounding-box analysis estimates relative distance and direction [5]. A priority classification module categorizes objects into high, medium, or low priority based on safety relevance. The silence module suppresses repetitive alerts [9]. When no immediate hazards are detected, the conditional OCR module [3] extracts visible text from the environment. Finally, processed information is converted to natural speech via an offline text-to-speech engine [6].

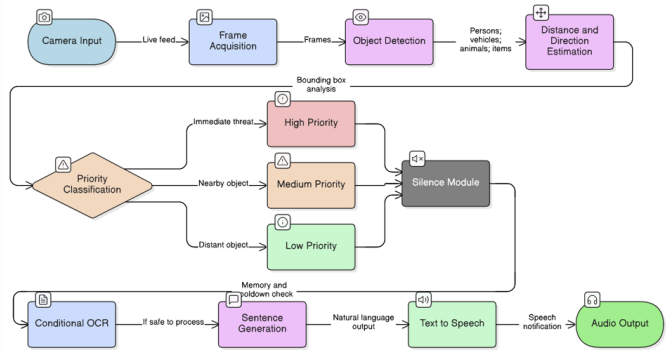

Fig. 1. System architecture of the VocalEyes context-aware assistive vision system.

B. Object Detection Methodology

The object detection module is realized using YOLOv8 [1], [10], a single-stage deep learning model that performs fast and accurate inference suitable for real-time assistive applications. The processing pipeline includes: (1) frame capture from the camera; (2) pre-processing (resizing and normalization); (3) inference through the deep learning model for object classification and bounding-box determination; and (4) labeling of objects with bounding-box coordinates. These bounding-box parameters form the basis for subsequent spatial estimation. In addition to the standard COCO-pretrained model, a custom YOLOv8 model is fine-tuned on a purpose-built dataset of approximately 2,500 annotated images covering stairs, doors, and text signs—object categories critical for blind user navigation but underrepresented in general datasets [9].

Fig. 2 illustrates an example where the system detects a road sign containing textual information. When no safety-critical objects are present, the OCR module [3] is activated to extract

textual content, enabling visually impaired users to receive environmental information such as traffic signs, directions, or notices through audio feedback.

Fig. 2. Example of object detection and text recognition in VocalEyes where a road sign is detected and processed for text extraction.

C. Distance and Direction Estimation

Where no depth sensing equipment is available [5], bounding-box width is used as a proxy for distance, based on the assumption that wider boxes represent closer objects. Distance is discretized into three categories: *near*, *medium*, and *far*. Direction is determined from the horizontal centroid of the bounding box relative to the frame width, yielding *left*, *center*, or *right*. This lightweight approach provides spatial awareness without additional instrumentation, directly addressing the sensor-dependency barrier identified in prior work [5], [8].

D. Priority-Based Classification

Detected objects are ranked into three priority levels:

- **High Priority:** vehicles, staircases, and moving hazards;
- **Medium Priority:** pedestrians and other mobile objects within proximity;
- **Low Priority:** non-portable objects such as furniture.

These levels determine the urgency of alert generation, whether a notification is issued immediately, conditionally, or suppressed, thereby ensuring safety-focused communication. Unlike systems that treat all detections equally [1], [4], [10], VocalEyes applies a hierarchical urgency model. The priority alert mechanism is illustrated in Fig. 3.

E. Intelligent Silence Mechanism

To counteract repetitive auditory output identified as a key usability problem in prior work [9], a memory-based suppression module is employed. The system retains: (1) the previously announced object label; (2) the associated distance category; and (3) the directional heading. Speech tokens are generated only when at least one of the following conditions is satisfied: a change in distance category or direction, the expiry of a cooldown timer, or the emergence of a new high-priority object. This approach minimizes cognitive overload [4] and improves user experience compared to continuous narration systems [9].

F. OCR Methodology

OCR is used to extract readable information from frames, extending the system's utility beyond obstacle detection [3]. The processing sequence includes: (1) frame pre-processing

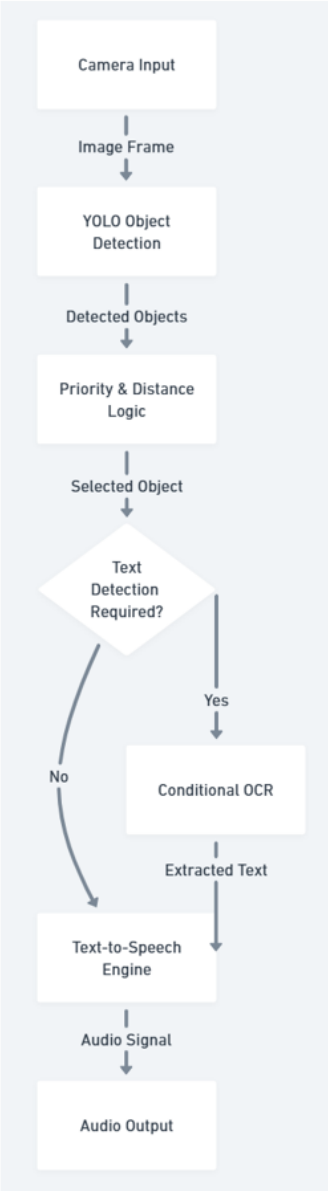

Fig. 3. Priority-based alert mechanism of VocalEyes illustrating the decision flow from object detection to conditional OCR activation and speech output.

(grayscale conversion, noise removal); (2) enhancement of textual areas; (3) OCR extraction; (4) validation of text length; and (5) application of a cooldown period to avoid unnecessary reads. Critically, OCR activation is conditional: when a high-priority object is detected, OCR is temporarily disabled to maintain focus on safety notifications. This directly addresses the uncontrolled OCR problem reported in existing reading-aid systems [3].

G. Speech Output Mechanism

Text-to-speech synthesis is performed using a native offline engine [6], eliminating the internet dependency and latency issues of cloud-based solutions. Decision logic ensures that only relevant, prioritized output triggers speech, enabling the

system to deliver meaningful and contextually appropriate guidance rather than fragmented detection labels as seen in earlier detect-and-speak approaches [1], [10].

H. Methodological Innovation

Unlike traditional assistive systems [1], [4], [9], [10] that follow a linear detection-to-speech pipeline, VocalEyes introduces a deliberative decision layer capable of: (1) assessing environmental hazards; (2) prioritizing alerts; (3) suppressing unnecessary speech; and (4) dynamically modulating OCR. This paradigm shift transforms a perception-based architecture into an advanced, decision-based assistive system that addresses the core limitations identified across the surveyed literature [1], [3]–[6], [8]–[10].

VI. Future Directions

The current implementation of VocalEyes provides real-time object recognition, spatial awareness, intelligent audio guidance, and safety-controlled OCR. Future enhancements may include:

- GPS-based navigation integrated with digital cartography for turn-by-turn audio guidance and landmark identification [6].
- Depth sensor or LiDAR integration to improve distance estimation in complex environments [5].
- Face recognition to identify known individuals and enhance social interaction [7].
- Multilingual speech support to increase accessibility across linguistically diverse populations.
- Emotion and gesture recognition for richer perception of the human environment.
- Deployment on wearable edge devices such as Raspberry Pi, Jetson Nano, or smart glasses [2], [7].
- Automated emergency detection to notify caregivers or emergency services during falls or unusual events.
- Optional cloud access and federated learning for periodic model updates across distributed devices.
- Integration with smart city infrastructure, including smart traffic lights and indoor beacons.
- Enhanced scene understanding and advanced OCR for structured document reading (menus, forms, information boards) [3].

VII. Conclusion

This paper presented VocalEyes, a context-aware assistive vision system designed to decode real-time visual data into intelligible audio guidance for visually impaired users. Unlike traditional assistive systems that rely on crude object detection and continuous speech [1], [9], [10], VocalEyes employs a decision-driven architecture integrating object detection, spatial estimation, priority-driven alerting, intelligent silence, and safety-constrained OCR.

The system's bounding-box–based spatial cognition estimates relative distance and direction without additional sensors [5]. A hierarchical classification mechanism ensures that safety-critical entities are reported promptly, while peripheral information is appropriately attenuated. The intelligent silence module reduces repetitive announcements [9], contributing to usability and reduced cognitive load [4]. The conditional OCR module [3] enables environmental text transcription but dynamically suppresses text reading when safety alerts take precedence.

VocalEyes features a modular, offline-first design [6] deployable without cloud connectivity on inexpensive computational substrates. The platform supports real-camera modalities for testing and practical implementation. Experimental evaluations confirm that the system provides real-time auditory feedback with minimal latency and reliable object recognition and classification [1], [10].

Beyond assistive navigation, the approaches described—real-time perception, priority-driven decision making, and multimodal data interpretation—have potential applications in intelligent surveillance and forensic investigation systems. VocalEyes demonstrates how the convergence of AI and context-driven decision logic can advance beyond simplistic detection-based paradigms [1], [4] toward intelligent, human-centered assistive technology, with considerable potential to improve the autonomy, safety, and quality of life of individuals with visual impairments.

Acknowledgment

The authors thank Dr. P. Renukadevi, Assistant Professor, Department of CSE-AIML, Jain University, for her guidance and support throughout this project.

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